

INNOVA JUNIOR COLLEGE
JC 2 PRELIMINARY EXAMINATION 2
in preparation for General Certificate of Education Advanced Level
Higher 1

CANDIDATE
NAME

CLASS

INDEX NUMBER

CHEMISTRY

8872/01

Paper 1 Multiple Choice

14 September 2011

50 minutes

Additional Materials: Data Booklet
 Multiple Choice Answer Sheet

READ THESE INSTRUCTIONS FIRST

Write your name and class on all the work you hand in.
Write in soft pencil.
Do not use staples, paper clips, highlighters, glue or correction fluid.

There are **thirty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A, B, C** and **D**.
Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.
Any rough working should be done in this booklet.

This document consists of **11** printed pages and **1** blank page.



1 Chromium is obtained from the following reaction, $2\text{Al} + \text{Cr}_2\text{O}_3 \longrightarrow 2\text{Cr} + \text{Al}_2\text{O}_3$. What is the percentage yield of chromium when 180 g of chromium is obtained from a reaction between 100 g of aluminium and 400 g of chromium(III) oxide?

- 2 0.5 g of zinc powder was found to reduce an acidified solution of 25.50 cm^3 of $0.200 \text{ mol dm}^{-3} \text{AO}_2^+$. What is the final oxidation number of metal **A**?

- 3 In an experiment, a sample of Americium element (Am) was vaporised, ionised and passed through an electric field. Analysis revealed that a beam of $^{241}\text{Am}^+$ gives an angle of deflection of 2° . What would be the angle of deflection for a sample of singly charged calcium ions?

- 4** The use of the *Data Booklet* is relevant to this question.

	1 st I.E.	2 nd I.E.
A	E < G	E > G
B	E < G	E < G
C	E > G	E > G
D	E > G	E < G

- 5** The successive ionisation energies (I.E.) of two elements, **J** and **L**, are given below.

I.E./ kJ mol ⁻¹	1 st	2 nd	3 rd	4 th	5 th	6 th	7 th	8 th
J	1090	2350	4610	6220	37800	47000	-	-
L	1251	2298	3822	5158	6542	9362	11018	33604

What is the likely formula of the compound that is formed when **J** reacts with **L**?

- | | | | |
|----------|-----------------------------------|----------|-----------------------|
| A | JL | C | JL₄ |
| B | J₂L₃ | D | J₄L |

- 6 Three substances Q, R and T have physical properties as shown.

Substance	Melting point/°C	Boiling point/°C	Electrical Conductivity	
			of solid	of liquid
Q	801	1413	Poor	Good
R	2852	3600	Poor	Good
T	3550	4827	Good	Unknown

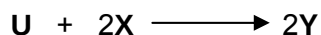
What could be the identities of **Q**, **R** and **T**?

	Q	R	T
A	$AlCl_3$	$NaCl$	C (diamond)
B	$AlCl_3$	MgO	C (graphite)
C	$NaCl$	MgO	C (graphite)
D	$NaCl$	MgO	C (diamond)

- 7 Which one of the following pairs contains substances with similar shape?

- A** CO_2 and Cl_2
B NH_3 and NO_3^-
C BeCl_2 and SnCl_2
D BCl_3 and PCl_3

- 8 The reaction between **U** and **X** is as follows:

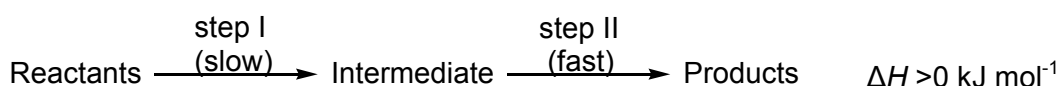


In an experiment to investigate the effect on the concentrations on rate of reaction, the following results were obtained at constant temperature.

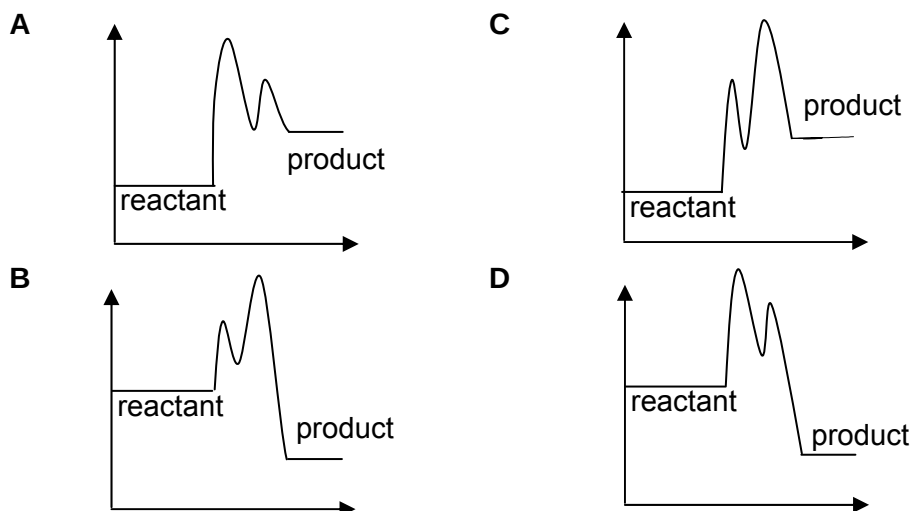
Experiment	[U]/ mol dm ⁻³	[X]/ mol dm ⁻³	Initial rate/ mol dm ⁻³ s ⁻¹
1	1.0	1.0	0.0008
2	1.0	2.0	0.0016
3	1.0	3.0	Z
4	2.0	2.0	0.0032

What is the value of **Z**?

- A 0.0008
 B 0.0016
 C 0.0024
 D 0.0032
- 9 The following reaction proceeds by two stages:

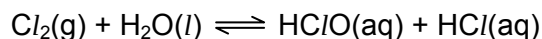


Which one of the following reaction pathway diagram correctly represents the energy changes in the above reaction?



- 10 Which of the following statements is always **true**?
- A At a given temperature, all molecules possess the same kinetic energy.
 B Halving the pressure of a gaseous reaction increases the rate of reaction.
 C The rate of a reaction increases as the reaction proceeds.
 D A catalyst increases the rates of both forward and backward reactions for a reversible reaction.

- 11 When chlorine gas dissolves in water, a solution smelling of Cl_2 is produced:



Which of the following will remove the smell of Cl_2 ?

- A addition of concentrated HCl to the solution
- B addition of NaOH to the solution
- C decreasing the pressure of the system
- D adding NaCl and allowing it to react with water
- 12 Which of the following statements regarding enthalpy change is **true**?
- A $\Delta H_{\text{rxn}}^\theta$ is the same regardless of the reaction pathway taken as long as the initial and final conditions are the same.
- B ΔH_c^θ is the energy change when one mole of the substance is burnt in air at 1 atm and 298 K.
- C ΔH_n^θ is the amount of energy released when 1 mole of acid reacts with 1 mole of base at 1 atm and 298 K.
- D ΔH_f^θ is the energy change when 1 mole of substance is formed from its constituent elements at 1 atm and 293 K.
- 13 The standard enthalpy change of the following reaction is $-896.4 \text{ kJ mol}^{-1}$.



	$\text{NO}(\text{g})$	$\text{H}_2\text{O}_2(\text{l})$
$\Delta H_f^\theta / \text{kJ mol}^{-1}$	+90.3	-187.8

Using the given information in the above table, what is the standard enthalpy change of formation, in kJ mol^{-1} , of $\text{HN}_3(\text{l})$?

- A +264
- B +528
- C +618
- D +632

- 14 The table shows the enthalpy change of neutralisation per mole of water formed, ΔH , for various acids and bases.

Acid	Base	$\Delta H / \text{kJ mol}^{-1}$
hydrochloric acid	sodium hydroxide	-57.0
E	sodium hydroxide	-54.0
hydrochloric acid	F	-52.0
nitric acid	G	-57.0

What are **E**, **F** and **G**?

	E	F	G
A	ethanoic acid	ammonia	potassium hydroxide
B	ethanoic acid	sodium hydroxide	ammonia
C	sulfuric acid	ammonia	potassium hydroxide
D	sulfuric acid	sodium hydroxide	ammonia

- 15 Given the following data, what is the standard enthalpy change of combustion of methanol, CH_3OH , in kJ mol^{-1} ?

$$\Delta H_c (\text{graphite}) = -394 \text{ kJ mol}^{-1}$$

$$\Delta H_f (\text{water}) = -286 \text{ kJ mol}^{-1}$$

$$\Delta H_f (\text{methanol}) = -239 \text{ kJ mol}^{-1}$$

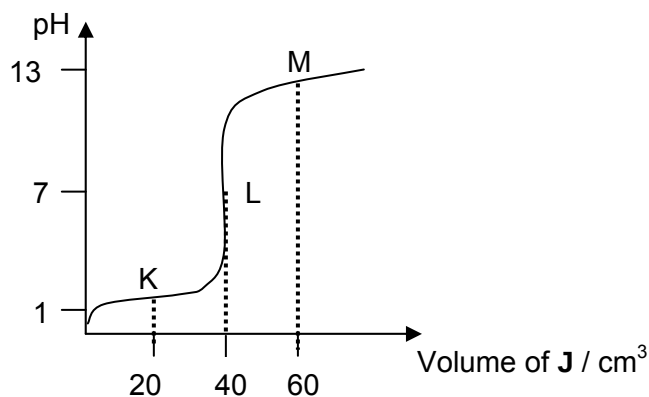
- | | | | |
|----------|------|----------|-------|
| A | -441 | C | -919 |
| B | -727 | D | -1025 |

- 16 An enzyme, found in stomach, operates at maximum efficiency when in an aqueous solution buffered at pH 5.

Which combination of substances when dissolved in 10 dm^3 of water, would give the necessary buffer solution?

- A** 1 mol HCl and 1 mol of CH_3COOH
B 1 mol CH_3COOH and 1 mol of CH_3COONa
C 1 mol HCl and 1 mol of CH_3COONa
D 1 mol of $\text{CH}_3\text{COONH}_4$ and 1 mole of NH_3

- 17 The following graph shows the pH changes when a 0.10 mol dm^{-3} solution of **J** is added to 20.0 cm^3 of 0.10 mol dm^{-3} of sulfuric acid.



Which of the following statements regarding the titration is **correct**?

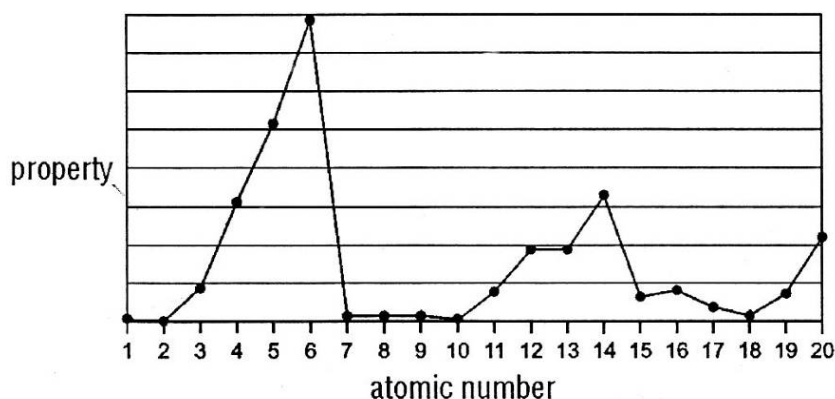
- A It is a weak monoprotic base.
 - B At point L, the salt at end point is hydrolysed to give an acidic solution.
 - C Phenolphthalein is a suitable indicator for this titration, but not methyl orange.
 - D There are no buffer regions in this graph.
- 18 The value of the ionic product of water, K_w , varies with temperature.

Temperature / °C	$K_w / \text{mol}^2 \text{ dm}^{-6}$
25	1.0×10^{-14}
62	1.0×10^{-13}

What can be deduced from this information?

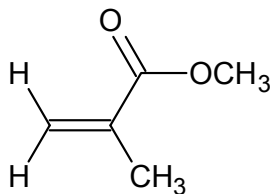
- A The ionic dissociation of water is an exothermic process.
 - B At 62°C , water with a pH of 6.5 is neutral.
 - C The pH of pure water increases with temperature.
 - D The association of water molecules by hydrogen bonding increases as temperature increases.
- 19 Consider the sequence of oxides Na_2O , SiO_2 , P_4O_{10} .
Which factor decreases from Na_2O to SiO_2 and also from SiO_2 to P_4O_{10} ?
- A covalent character
 - B electrical conductivity in the solid state
 - C pH when mixed with water
 - D solubility in aqueous alkali

- 20 The following shows the variation of a property of the first 20 elements in the Periodic Table with the atomic number of the element.



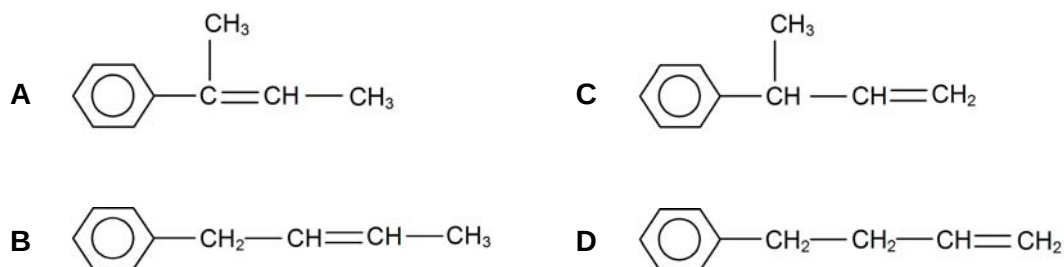
What is the property?

- A Atomic radius
B First ionisation energy
C Ionic radius
D Melting point
- 21 Which of the following statements is **true** of the organic compound shown below?



- A It is a cis isomer.
B It has three sp^2 carbon atoms and two sp^3 carbon atoms.
C It has 7σ and 2π bonds.
D It will form a ketone as one of the products with hot acidified $K_2Cr_2O_7(aq)$.
- 22 Compound E
- has the molecular formula $C_{10}H_{14}O$;
 - is unreactive towards mild oxidising agents.

What is the structure of the compound formed by dehydration of E?



- 23** The table shows the results of simple tests on compound **G**.

Reagent	Result
2,4-dinitrophenylhydrazine	Orange precipitate
Ammoniacal silver nitrate	Silver mirror

From the results of the tests, what could **G** be?

- A** $\text{CH}_3\text{CH}_2\text{COCH}_3$ **C** $\text{CH}_3\text{CH}_2\text{CH}_2\text{CHO}$
B $\text{CH}_3\text{CH}(\text{OH})\text{CH}_3$ **D** $\text{CH}_3\text{CH}_2\text{CH}_2\text{COOH}$

- 24** Which compound is a product of hydrolysis of $\text{CH}_3\text{CO}_2\text{C}_3\text{H}_7$ by boiling hydrochloric acid?

- A** $\text{CH}_3\text{COC}l$
B $\text{C}_3\text{H}_7\text{CO}_2\text{H}$
C $\text{C}_3\text{H}_7\text{COC}l$
D $\text{C}_3\text{H}_7\text{OH}$

- 25** A student attempted to perform a reaction on 3 different organic compounds using different reducing agents. He recorded his results as shown in the table below:

Test	Compounds	H ₂ /Ni	LiAlH ₄	NaBH ₄
I	CH ₃ CH ₂ COCH ₃	√	√	√
II	CH ₃ CH=CHCH ₃	√	X	X
III	CH ₃ CH ₂ COOH	X	√	X

Which of the test results are correct?

- A** I & II **C** II & III
B I & III **D** I, II & III

Section B

For each of the questions in this section, one or more of the three numbered statements **1** to **3** may be correct.

Decide whether each of the statements is or is not correct (you may find it helpful to put a tick against the statements that you consider to be correct).

The responses **A** to **D** should be selected on the basis of

A	B	C	D
1, 2 and 3 are correct	1 and 2 are correct	2 and 3 are correct	1 only is correct.

- 26** Which statement(s) about a 12.0 g sample of ^{12}C are correct?
- The number of atoms in the sample is 6.02×10^{23} .
 - The number of atoms in the sample is the same as the number of atoms in 4.0 g of helium.
 - The number of atoms in the sample is the same as the number of atoms in 28.0 g of nitrogen gas.
- 27** Which of the following option(s) **must** be true of a weak acid which is relatively strong?
- It has a high K_a value.
 - It is an acid with high concentration.
 - Its conjugate base is strong.
- 28** Which of the following trend(s) concerning Period 3 elements from Na to Cl are **true**?
- There is a change from metallic behaviour to non-metallic behaviour.
 - Their compounds show an increase in the maximum oxidation number across the period.
 - The melting points of the elements decrease across the period.
- 29** In an experiment, the complete combustion of a hydrocarbon yields 3.52 g of carbon dioxide and 1.44 g of water. Assume that all measurements were obtained at same temperature and pressure.

Which of the following molecular formula(e) would fit these data?

- C_2H_4
- C_4H_8
- C_6H_6

- 30 Which of the following compound(s) will form potassium benzoate, $\text{C}_6\text{H}_5\text{COO}^-\text{K}^+$ when refluxed in the presence of aqueous potassium hydroxide?

